

Mustafa Sultan

AI Engineer | Agentic AI & LLM Applications | ML/DL & Data Science

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Professional Summary

AI Engineer and recent Artificial Intelligence graduate with hands-on experience in machine learning, LLM applications, agentic workflows, RAG, backend APIs, data science, and computer vision. Built practical AI systems through internships, research, and academic projects, and contributed to a peer-reviewed publication.

Technical Skills

Programming:	Python, SQL, C++, C#
AI/ML:	PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, Optuna, Machine Learning, Deep Learning, NLP, Computer Vision, Medical Imaging
Generative AI:	LLM Integration, LangGraph, LangChain, Agentic AI, Multi-Agent Systems, RAG, Hybrid Search, Reranking, Embeddings, Retrieval, Prompt Engineering, Tool Calling
Backend & Data:	Django, Django Ninja, FastAPI, REST APIs, PostgreSQL, SQL Databases, Pandas, NumPy, Pydantic, API Integration
Software Engineering:	Git, GitHub, OOP, Data Structures and Algorithms, Version Control
Additional:	Qiskit, Quantum Optimization, Competitive Programming

Education

B.Sc. in Artificial Intelligence

Oct 2022 – Jun 2026

Nile University

Egypt

- GPA: 3.82 / 4.00; Dean's List: Fall 2024.
- Completed and passed all academic requirements; official graduation documentation pending.

Experience

AI Engineer Intern

Jul 2025 – Sep 2025

Crossworkers Egypt

Cairo, Egypt

- Built an AI lead-generation pipeline that analyzed email newsletters to extract job postings and company information.
- Developed a machine learning model to predict employee satisfaction from historical HR data.
- Created an HR chatbot for natural-language querying over SQL databases.
- Integrated LLM APIs and LangGraph to orchestrate agentic workflows across backend services and live databases.
- Implemented application components using Python and C# for internal AI workflows.

Research Intern - Machine Learning Classification

Jun 2024 – Sep 2024

Nile University

Giza, Egypt

- Performed preprocessing, EDA, and feature engineering on mobile traffic and POI datasets.
- Implemented and evaluated classification models for socioeconomic indicator prediction.
- Contributed to a peer-reviewed publication based on the project results.

Research Intern - Quantum Optimization

Jun 2024 – Sep 2024

Nile University

Giza, Egypt

- Applied QAOA to unit commitment optimization problems and explored quantum optimization concepts in a research setting.

Selected Projects

FGaiB - Multi-Agent Personal Finance Assistant

2025 – 2026

- Designed Django Ninja REST APIs and PostgreSQL data models for transactions, budgets, savings goals, recurring bills, notifications, and assistant-driven actions.
- Built a LangGraph orchestration workflow that routes requests to specialized financial agents and maintains relevant user context across interactions.
- Integrated tool-using LLM logic with structured financial operations, schema validation, conversation memory, and confirmation flows before modifying user records.
- Supported proactive financial insights, including budget-risk alerts, goal-progress monitoring, spending analysis, and natural-language interaction with financial data.
- The project ranked first in one of eight faculty graduation project committees.

SlideTutor - AI-Powered PowerPoint Study Workspace

github.com/mustafa200312/slidetutor

- Built a full-stack educational AI application that extracts and renders PowerPoint slides for interactive study.
- Developed a FastAPI backend and React interface for slide content, images, structured elements, and slide-aware chat.
- Implemented LangGraph summarizer and interactive agents with conversation memory and OpenAI-compatible model support.

Al-Maktaba Al-Digital - Arabic Book-Constrained RAG System

- Built a citation-enforced RAG system over selected Arabic classical books and texts.
- Implemented Arabic document processing, chunking, embeddings, hybrid search, reranking, and grounded answer generation.
- Combined semantic and keyword-based retrieval, then reranked retrieved passages to improve source relevance before answer generation.

Multi-View Ultrasound Medical Imaging with a Vision Foundation Model

- Developed a two-view PyTorch medical imaging model using a frozen pretrained DINOv3 vision foundation model to analyze longitudinal and transverse ultrasound scans.
- Built three-class segmentation heads to identify vessel and plaque regions and a binary classification head that combines information from both views for plaque detection.
- Adapted grayscale ultrasound images for pretrained vision features and jointly trained the segmentation and classification tasks using cross-entropy and Dice-based losses.
- Addressed class imbalance with weighted sampling and evaluated performance using per-class Dice and F1 score.

MAAP - Multi-Agent System for Automatic Code Parallelization

github.com/mustafa200312/MAAP-Multi-Agent-System-for-Auto-Parallelization

- Built a multi-agent LLM system that transforms sequential Python and C code into parallel implementations using agents for dependency detection, task decomposition, and execution planning.

Publication & Achievements

- *A Multi-label Classification Framework for Socioeconomic Insights Using Mobile Traffic and POI Data.* doi.org/10.1007/978-3-031-87719-3_16
- Qualified for ECPC 2023 and 2024.
- Won 2nd Place in the Linear Algebra Course category at the 17th Undergraduate Research Forum, Nile University.

Volunteer Experience

Junior Teaching Assistant, Nile University

Oct 2023 – Jan 2024

Supported students in programming, data structures, data analysis, algorithms, preprocessing, and feature engineering.

ICPC Community Mentor, Nile University

Feb 2024 – Jun 2024

Mentored peers in competitive programming and algorithmic problem solving.